

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione ([filippo dot castiglione at cnr dot it](mailto:filippo.castiglione@cnr.it))

Technical problems: Ilaria Gonnella ([ilaria dot gonnella at cnr dot it](mailto:ilaria.gonnella@cnr.it))

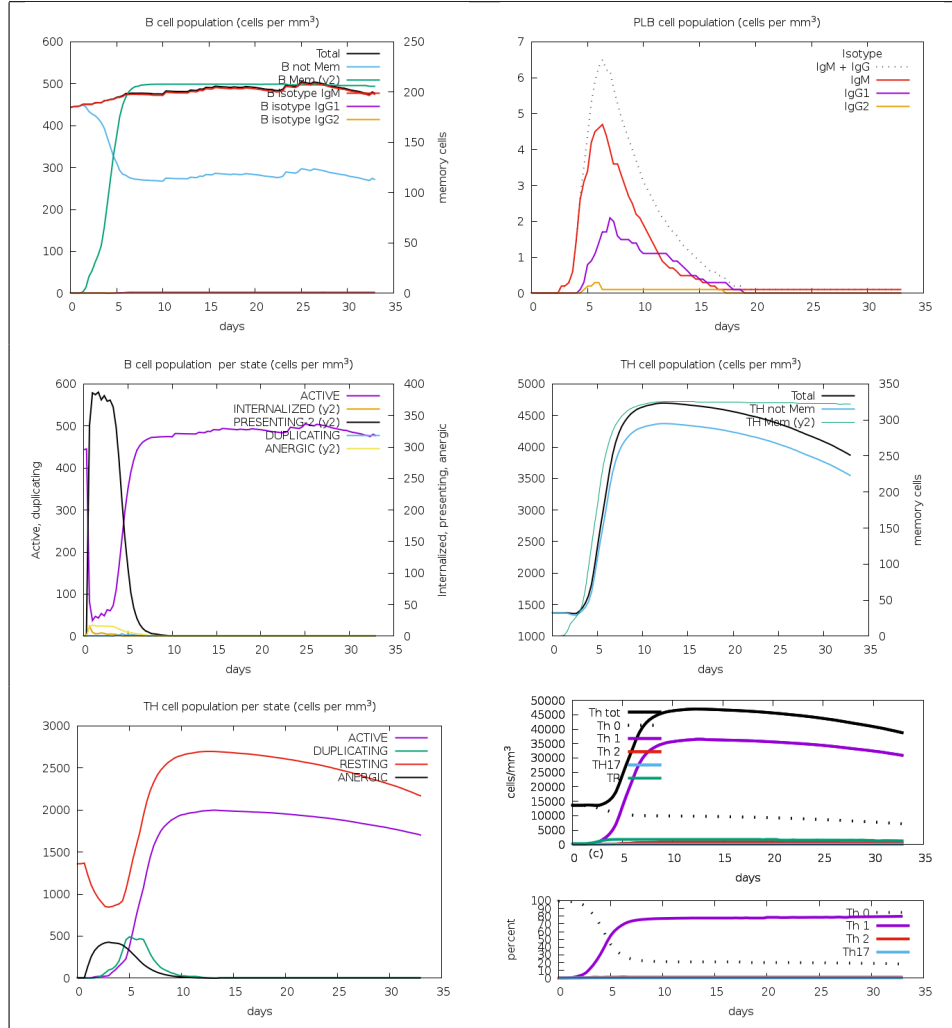


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

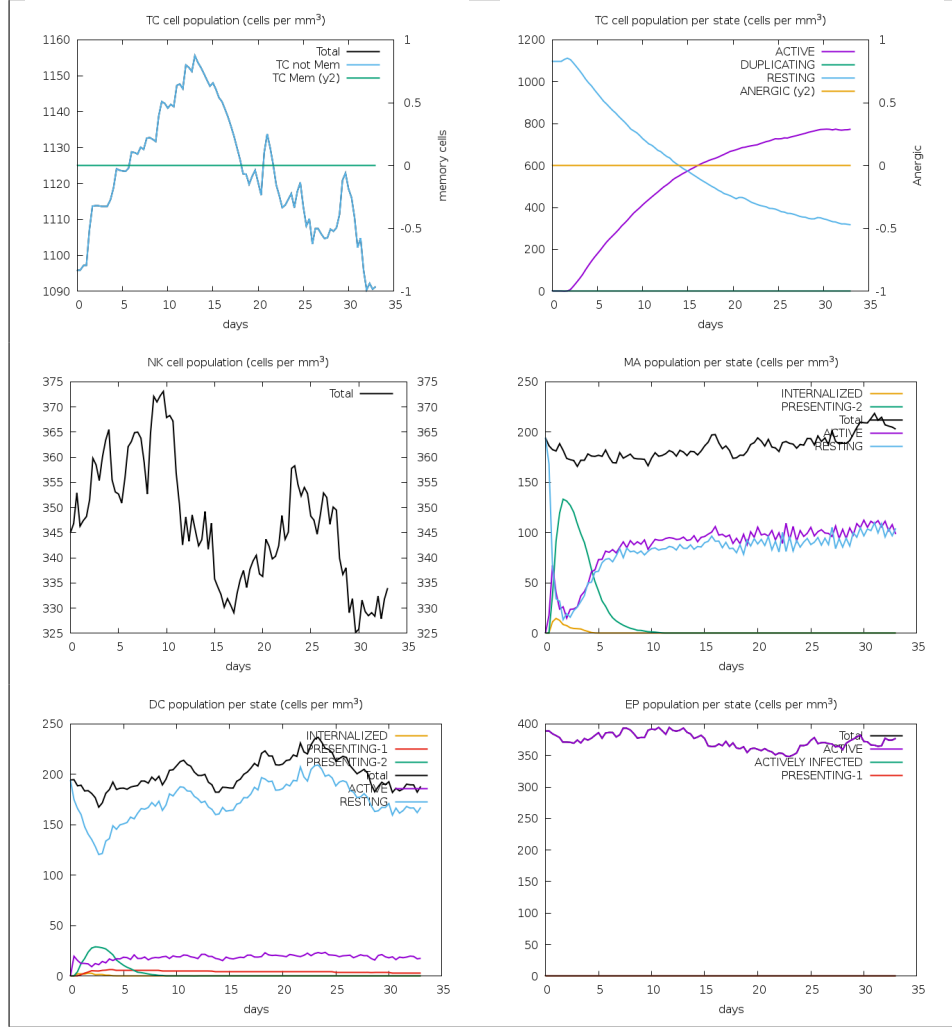


Figure 2: Legend: symbols as figure above.

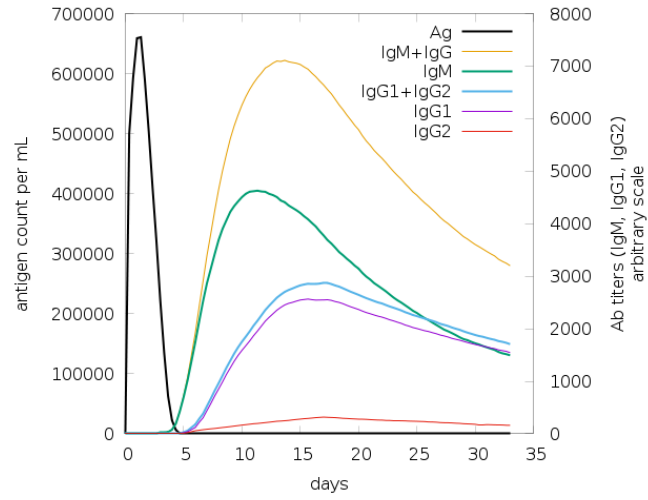


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

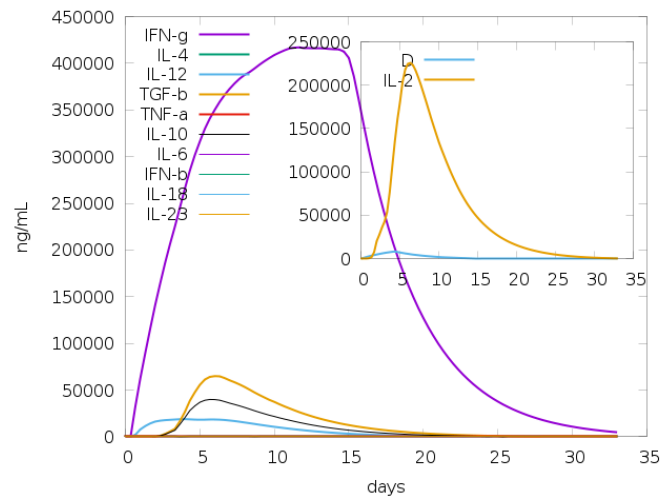


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Parker_B_scale_epitope_find

Use Parker's propensity scale, takes an antigen block as input,
and creates a list of residues that are possible epitopes.

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014305_5_AmVGOLKldjZg.FSA_1_001

AKFVAAWTLKAAAGGSYTIPWTELMAAAAVTVAFWAWAAAAESLWVGVRAAAAIFLIGFALFAAAAANVFDLEGIF
RAAAAAKLPHWPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTAAAAAALAEVDIPAVAAAAASDREVRGSALYEAST
AAAAATSSSVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAPPMVEGAAAEGDDGD
AAAAAVFSDEIPASQDMPTNTAAAAADGKNKETFERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
GGTFK

0000000000001000
00
00
1000
100000000000110011110000001111101000
00100

Epitopes of protein 0 -----

1]	pos=113 len=4	AGTD
2]	pos=144 len=7	ASDREVR
3]	pos=232 len=9	AAEGDDGDA
4]	pos=255 len=4	DMPT
5]	pos=265 len=6	ADGKNK

DoPeptideList_I:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

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Allele: A0101

Pseudo sequence: KAVHAEQRNKAQTRA

Threshold: 9.456400

Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014305_5_AmVGOLKldjZg.FSA_1_001

AKFVAAWTLKAAAGGSYTIPWTELMAAAAVTVAFWAWAAAAESLWVGVRAAAAIFLIGFALFAAAAANVFDLEGIF
RAAAAAKLPHWPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTAAAAAALAEVDIPAVAAAAASDREVRGSALYEAST
AAAAATSSSVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAPPMVEGAAAEGDDGD
AAAAAVFSDEIPASQDMPTNTAAAAADGKNKETFERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
GGTFK

Epitopes of protein 0 -----

0]	pos= -1 score=1.000000 unnormalised=104.8410000000	non-binding event
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Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014305_5_AmVGOLKldjZg.FSA_1_001

AKFVAAWTLKAAAGGSYTIPWTELMAAAAAVTVTAFWAAAAAAESLWVGVRAAAAIFLIGFALFAAAAAANVFDLEGIF
RAAAAAAKLPHWPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTAAAAAALAEVDIPAVAAAAASDREVRGSALYEAST
AAAAATSSFVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAAPPMVEGAAAEAGDDGD
AAAAAVFSDEIPASQDMPTNTAAAAADGKNKETFERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
GGTFK

Epitopes of protein 0 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

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Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014305_5_AmVGOLKldjZg.FSA_1_001

AKFVAAWTLKAAAGGSYTIPWTELMAAAAAVTVTAFWAAAAAAESLWVGVRAAAAIFLIGFALFAAAAAANVFDLEGIF
RAAAAAAKLPHWPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTAAAAAALAEVDIPAVAAAAASDREVRGSALYEAST
AAAAATSSFVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAAPPMVEGAAAEAGDDGD
AAAAAVFSDEIPASQDMPTNTAAAAADGKNKETFERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
GGTFK

Epitopes of protein 0 -----

0]	pos=	87	score=0.008238	unnormalised=1.0912000000	LPHWPTLA
1]	pos=	91	score=0.026840	unnormalised=3.5552000000	TPTLAAAAA
2]	pos=	102	score=0.006192	unnormalised=0.8202000000	IPEFPFYAA
3]	pos=	105	score=0.037084	unnormalised=4.9122000000	FPFYAAAAA
4]	pos=	136	score=0.037892	unnormalised=5.0192000000	IPAVAAAAA
5]	pos=	160	score=0.013621	unnormalised=1.8042000000	AAAAATSSF
6]	pos=	215	score=0.018875	unnormalised=2.5002000000	IPNLFAAAA
7]	pos=	219	score=0.003316	unnormalised=0.4392000000	FAAAAAAPM
8]	pos=	224	score=0.000779	unnormalised=0.1032000000	APPMVEGAA
9]	pos=	250	score=0.015372	unnormalised=2.0362000000	IPASQDMPT
10]	pos=	256	score=0.040308	unnormalised=5.3392000000	MPTNTAAAA
11]	pos=	-1	score=0.791485	unnormalised=104.8410000000	non-binding event

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Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014305_5_AmVGOLKldjZg.FSA_1_001

AKFVAAWTLKAAAGGSYTIPWTELMAAAAAVTVTAFWAAAAAAESLWVGVRAAAAIFLIGFALFAAAAAANVFDLEGIF
RAAAAAAKLPHWPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTAAAAAALAEVDIPAVAAAAASDREVRGSALYEAST
AAAAATSSFVTNTTVGIELPAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAAPPMVEGAAAEAGDDGD

AAAAVFSDEIPASQDMPTNTAAAAADGKNKETFHERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
GGTFK

Epitopes of protein 0 -----

0]	pos=	87	score=0.008238	unnormalised=1.0912000000	LPHWTPTLA
1]	pos=	91	score=0.026840	unnormalised=3.5552000000	TPTLAAAAA
2]	pos=	102	score=0.006192	unnormalised=0.8202000000	IPEFPFYAA
3]	pos=	105	score=0.037084	unnormalised=4.9122000000	FPFYAAAAA
4]	pos=	136	score=0.037892	unnormalised=5.0192000000	IPAVAAAAA
5]	pos=	160	score=0.013621	unnormalised=1.8042000000	AAAAATSSF
6]	pos=	215	score=0.018875	unnormalised=2.5002000000	IPNLFAAAA
7]	pos=	219	score=0.003316	unnormalised=0.4392000000	FAAAAAAPPM
8]	pos=	224	score=0.000779	unnormalised=0.1032000000	APPMVEGAA
9]	pos=	250	score=0.015372	unnormalised=2.0362000000	IPASQDMPT
10]	pos=	256	score=0.040308	unnormalised=5.3392000000	MPTNTAAAA
11]	pos=	-1	score=0.791485	unnormalised=104.8410000000	non-binding event

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014305_5_AmVGOLKldjZg.FSA_1_001

AKFVAAWTLKAAAGGSYTIPTWELMAAAAVTVTAFWAWAAAAAESLWVGVRAAAAIFLIGFALFAAAAAANVFDLEGIF
RAAAAAKLPHTPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTKAAAAALAEVDIPAVAAAAASDREVRGSALYEAST
AAAAATSSFVNTTGVIELPAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAPPMVEGAAAEGLDGD
AAAAVFSDEIPASQDMPTNTAAAAADGKNKETFHERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
GGTFK

Epitopes of protein 0 -----

0]	pos=	2	score=0.011091	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.001513	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	20	score=0.026235	unnormalised=8.0055600000	WTELMAAAA
3]	pos=	21	score=0.002902	unnormalised=0.8855600000	TELMAAAAA
4]	pos=	23	score=0.023964	unnormalised=7.3125600000	LMAAAAVTV
5]	pos=	24	score=0.025098	unnormalised=7.6585600000	MAAAAVTV
6]	pos=	35	score=0.014536	unnormalised=4.4355600000	FWAWAAAAA
7]	pos=	36	score=0.009741	unnormalised=2.9725600000	WAWAAAAAE
8]	pos=	38	score=0.035751	unnormalised=10.9095600000	WAAAAAESL
9]	pos=	50	score=0.003135	unnormalised=0.9565600000	VYRAAAAIIF
10]	pos=	51	score=0.039838	unnormalised=12.1565600000	YRAAAAIIFL
11]	pos=	58	score=0.009056	unnormalised=2.7635600000	FLIGFALFA
12]	pos=	62	score=0.028289	unnormalised=8.6325600000	FALFAAAAA
13]	pos=	64	score=0.021801	unnormalised=6.6525600000	LFAAAAANV
14]	pos=	65	score=0.018393	unnormalised=5.6125600000	FAAAAAANV
15]	pos=	78	score=0.005435	unnormalised=1.6585600000	IFRAAAAAK
16]	pos=	79	score=0.045009	unnormalised=13.7345600000	FRAAAAAKL
17]	pos=	90	score=0.023675	unnormalised=7.2245600000	WPTLAAAAA

18]	pos= 94	score=0.016869	unnormalised=5.1475600000	LAAAAAVEI
19]	pos= 107	score=0.037662	unnormalised=11.4925600000	FYAAAAAGT
20]	pos= 119	score=0.013333	unnormalised=4.0685600000	IFLTKAAAA
21]	pos= 120	score=0.024324	unnormalised=7.4225600000	FLTKEAAAA
22]	pos= 121	score=0.023253	unnormalised=7.0955600000	LTKAAAAAA
23]	pos= 122	score=0.000352	unnormalised=0.1075600000	TKAAAAAAL
24]	pos= 136	score=0.007084	unnormalised=2.1615600000	IPAVAAAAA
25]	pos= 154	score=0.003921	unnormalised=1.1965600000	LYEASTAAA
26]	pos= 155	score=0.030895	unnormalised=9.4275600000	YEASTAAAA
27]	pos= 168	score=0.013274	unnormalised=4.0505600000	FVTNTTVGI
28]	pos= 176	score=0.011875	unnormalised=3.6235600000	IELPAAAAA
29]	pos= 188	score=0.014067	unnormalised=4.2925600000	FNSILFALL
30]	pos= 192	score=0.033359	unnormalised=10.1795600000	LFALLAAAA
31]	pos= 193	score=0.049259	unnormalised=15.0315600000	FALLAAAAA
32]	pos= 195	score=0.031350	unnormalised=9.5665600000	LLAAAAAAA
33]	pos= 196	score=0.029512	unnormalised=9.0055600000	LAAAAAAA
34]	pos= 206	score=0.025668	unnormalised=7.8325600000	FKSLSSTEF
35]	pos= 215	score=0.000703	unnormalised=0.2145600000	IPNLFAAAA
36]	pos= 218	score=0.003148	unnormalised=0.9605600000	LFAAAAAPP
37]	pos= 219	score=0.043495	unnormalised=13.2725600000	FAAAAAAPM
38]	pos= 280	score=0.018298	unnormalised=5.5835600000	FAAAAAAPS
39]	pos= 301	score=0.005645	unnormalised=1.7225600000	VLHEDAAAA
40]	pos= 302	score=0.000356	unnormalised=0.1085600000	LHEDAAAAA
41]	pos= -1	score=0.216837	unnormalised=66.1680000000	non-binding event

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Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014305_5_AmVGOLKldjZg.FSA_1_001

AKFVAAWTLKAAAGGSYTIPTWELMAAAAVTVTAFWAAAAAAESLWVGVRAAAAIFLIGFALFAAAAAANVFDLEGIF
RAAAAAKLPHTPTLAAAAAVEIPEFPFYAAAAAGTDNEIFLTAAAAAALAEVPDIPAVAAAAASDREVRGSALYEAST
AAAAATSSFVTNTTVGIELPAAAAADPPFSILFALLAAAAAANFKSLSSTEFIPNLFAAAAAPPMEGAAAEGDDGD
AAAAAVFSDEIPASQDMPNTNTAAAAADGKNKETFERADSFAAAAACPSLDVDSIIPRTPDVLHEDAAAAAGEGNWTS
GGTFK

Epitopes of protein 0 -----

0]	pos= 2	score=0.011091	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001513	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 20	score=0.026235	unnormalised=8.0055600000	WTELMAAAA
3]	pos= 21	score=0.002902	unnormalised=0.8855600000	TELMAAAAA
4]	pos= 23	score=0.023964	unnormalised=7.3125600000	LMAAAAVT
5]	pos= 24	score=0.025098	unnormalised=7.6585600000	MAAAAVTV
6]	pos= 35	score=0.014536	unnormalised=4.4355600000	FWAAAAAA
7]	pos= 36	score=0.009741	unnormalised=2.9725600000	WAAAAAAE
8]	pos= 38	score=0.035751	unnormalised=10.9095600000	WAAAAAESL
9]	pos= 50	score=0.003135	unnormalised=0.9565600000	VYRAAAIF
10]	pos= 51	score=0.039838	unnormalised=12.1565600000	YRAAAIFL
11]	pos= 58	score=0.009056	unnormalised=2.7635600000	FLIGFALFA
12]	pos= 62	score=0.028289	unnormalised=8.6325600000	FALFAAAAA
13]	pos= 64	score=0.021801	unnormalised=6.6525600000	LFAAAAANV
14]	pos= 65	score=0.018393	unnormalised=5.6125600000	FAAAAAANV
15]	pos= 78	score=0.005435	unnormalised=1.6585600000	IFRAAAAAK
16]	pos= 79	score=0.045009	unnormalised=13.7345600000	FRAAAAAKL
17]	pos= 90	score=0.023675	unnormalised=7.2245600000	WTPTLAAAA
18]	pos= 94	score=0.016869	unnormalised=5.1475600000	LAAAAAVEI
19]	pos= 107	score=0.037662	unnormalised=11.4925600000	FYAAAAAGT
20]	pos= 119	score=0.013333	unnormalised=4.0685600000	IFLTKAAAA

21]	pos= 120	score=0.024324	unnormalised=7.4225600000	FLTKAAAAA
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23]	pos= 122	score=0.000352	unnormalised=0.1075600000	TKAAAAAAL
24]	pos= 136	score=0.007084	unnormalised=2.1615600000	IPAVAAAAA
25]	pos= 154	score=0.003921	unnormalised=1.1965600000	LYEASTAAA
26]	pos= 155	score=0.030895	unnormalised=9.4275600000	YEASTAAAA
27]	pos= 168	score=0.013274	unnormalised=4.0505600000	FVTNTTVGI
28]	pos= 176	score=0.011875	unnormalised=3.6235600000	IELPAAAAA
29]	pos= 188	score=0.014067	unnormalised=4.2925600000	FNSILFALL
30]	pos= 192	score=0.033359	unnormalised=10.1795600000	LFALLAAAA
31]	pos= 193	score=0.049259	unnormalised=15.0315600000	FALLAAAAA
32]	pos= 195	score=0.031350	unnormalised=9.5665600000	LLAAAAAAA
33]	pos= 196	score=0.029512	unnormalised=9.0055600000	LAAAAAAA
34]	pos= 206	score=0.025668	unnormalised=7.8325600000	FKSLSTEF
35]	pos= 215	score=0.000703	unnormalised=0.2145600000	IPNLFAAAA
36]	pos= 218	score=0.003148	unnormalised=0.9605600000	LFAAAAAPP
37]	pos= 219	score=0.043495	unnormalised=13.2725600000	FAAAAAAPP
38]	pos= 280	score=0.018298	unnormalised=5.5835600000	FAAAAAACPS
39]	pos= 301	score=0.005645	unnormalised=1.7225600000	VLHEDAAAA
40]	pos= 302	score=0.000356	unnormalised=0.1085600000	LHEDAAAAA
41]	pos= -1	score=0.216837	unnormalised=66.1680000000	non-binding event
